Ivar Cashin Eriksson

Curriculum Vitae

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Introduction

I am a data scientist and ML engineer with seven years of machine-learning experience and a strong quantitative background from KTH's Engineering Physics programme and MSc specialisation in Statistical Learning and Data Analytics. My work combines deep learning, NLP, embeddings, vector search, probabilistic modelling, and production ML ownership in both business and research settings. I am extremely excited to apply to kapa.ai because I want to move closer to ML research while building systems that work reliably in practice.

Education

2015–2020 **Royal Institute of Technology, KTH – Data Science and Engineering**, GPA: 5.0 (out of 5.0)
Master: Statistical Learning and Data Analytics, Bachelor: Engineering Physics. See thesis.
Relevant coursework: Statistical Machine Learning; Modern Methods of Statistical Learning.

2019–2019 **Swiss Federal Institute of Technology, ETH – Applied Mathematics**, *Exchange*

2017–2022 **Stockholm University, SU - Economics**, *Supplementary Bachelor's Degree*

Selected Project, Research, and ML Engineering Experience

Solo Project @ `resume_builder3`

- Building a source-grounded system for generating tailored application materials.
- Produces evidence mappings and gap reports to keep generated claims traceable to profile YAMLs.
- Runs job-hunter agent workflow from web crawling, to GitHub Actions, and automatic application materials.

Keywords: Agentic workflows, source-grounded generation, human-in-the-loop automation, evidence mapping, GitHub Actions, Codex

Read more...

Researcher within Data Science @ RaySearch Laboratories

- Learned latent anatomical representations from segmented 3D CT scans for cancer treatment automation workflows using autoencoders.
- Predicted dose-volume histograms and spatial dose distributions using sparse Gaussian processes.

Keywords: deep learning, representation learning, probabilistic modelling, sparse Gaussian processes, optimisation, medical AI

Read more...

Principal Data Scientist @ Valcon

- Architected, project managed, and built an in-production, real-time, deep neural network system using NLP, autoencoders, and transfer learning.
- Led client projects from problem framing through model delivery and stakeholder alignment.

Keywords: NLP, deep neural networks, production ML, explainable AI, stakeholder management, mentoring

Read more...

Data Scientist & Project Manager @ Signum Life Science

- Use LLMs to embed free text for downstream similarity search and matching workflows.
- Work as architect, data scientist, ML engineer, software engineer, and project manager.

Keywords: LLM embeddings, search, explainable AI, Databricks, XGBoost, project management

Read more...

Skills and Other

Technology	Python, SQL, PyTorch, TensorFlow, Scikit-learn, SHAP, OpenCLIP, MLflow, FastAPI, Docker, MongoDB, Qdrant, Databricks, AWS, MS Azure.
Methods	Deep learning, representation learning, embeddings, vector search, explainable AI.
Language	*Native*: English, Swedish. *Intermediate*: Danish.
Other	Valcon People's Choice Award (2023); IYPT Sweden (2015); Mensa Sweden; Woodworker.

Detailed Work Experience

2024–Present **Data Scientist and Project Manager**, *Signum Life Science*, Copenhagen

At Signum, I work across architecture, project management, data science, ML engineering, and software engineering for many ongoing projects. The role requires taking loosely defined business problems and turning them into maintainable technical work.

In my current task, I am using LLMs to embed free text for downstream similarity search and matching workflows. The work focuses on the retrieval layer of RAG-style systems: embedding free text, evaluating semantic similarity, analysing retrieval failures, and improving matching quality in domain-specific workflows.

Alongside this, I am working on a forecasting platform for pharmaceutical price and demand forecasting, including patient population forecasts. I work across architecture, project management, data science, ML engineering, and software engineering for the platform, with Databricks as a core part of the implementation. The platform is being designed to provide thousands of simultaneous forecasts generated by thousands of independently trained models.

Beyond model development, I organised an internal week-long hackathon with up to 40 participants.

2022–2024 **Principal Data Scientist**, *Valcon*, Copenhagen

At Valcon, I led data science projects across pharma, energy, IoT, and manufacturing. The role involved senior ownership in settings where scope, data quality, and stakeholder expectations were often not fully defined at the outset.

One project involved training a real-time deep neural network with a custom training pipeline and modular architecture. The training data consisted of natural-language inputs and high-dimensional multi-class labels, which made for a challenging prediction problem. I was responsible for the end-to-end production pipeline.

I mentored junior colleagues, led explainable AI and Git trainings, and regularly presented technical topics to non-technical stakeholders. I was awarded the Valcon People's Choice Award for exemplary work, leadership, team spirit, and scientific curiosity.

2021–2022 **Data Scientist**, *Valtech*, Copenhagen

2019–2021 **Researcher within Data Science**, *RaySearch Laboratories*, Stockholm

At RaySearch Laboratories, I worked on automating radiotherapy planning using deep learning, probabilistic modelling, and multi-objective optimisation. I developed a pipeline that extracted latent anatomical features from segmented 3D CT scans using autoencoders, then used those representations to identify similar patients via a learned similarity metric. Dose-volume histograms and spatial dose distributions were then predicted using sparse Gaussian processes, providing a probabilistic framework for dose planning conditioned on patient geometry. I also redesigned lexicographic optimisation algorithms to better reflect clinical decision hierarchies.

I collaborated closely with medical physicists and clinicians throughout, ensuring the models aligned with oncological standards and practical workflow requirements.

Selected Projects

`resume_builder3` Source-grounded agentic workflow for job applications. The system uses structured profile YAML, job listings, LaTeX templates, and previous examples to generate tailored CVs and cover letters. It also produces evidence mappings and gap reports so major claims can be traced back to source material and weak fit is visible before submission. I extended the workflow with a human-in-the-loop job-hunter agent that crawls the web for relevant adverts, sends a daily digest, waits for my response, creates GitHub issues for selected roles, triggers GitHub Actions/Codex to generate a branch with listing information and application materials, and opens a PR for review. This application is of course generated with this system!

Project `iris` Computer-vision retrieval system for identifying fashion items across web-shop images. Built object detection, embedding generation, vector indexing, and similarity search using YOLOv8, OpenCLIP, Qdrant, FastAPI, MongoDB, scraping, and a Chromium extension.